

Automated Podcast Transcription and Topic Segmentation

1. Overview

This project is an AI-powered application designed to streamline the consumption of audio content. It automates the process of transcribing podcast audio into text, segments the content based on topic shifts, and visualizes key themes. The tool allows users to upload audio files and receive a structured summary, complete with keyword insights and interactive visualizations, making it easier to navigate long-form audio content without listening to the entire file.

2. Approach

The application follows a linear, modular pipeline designed to convert raw unstructured audio into structured, actionable insights. The system architecture is divided into four distinct phases:

Phase 1: Audio Ingestion & Preprocessing

Before any AI processing, the raw audio file must be normalized to ensure compatibility with the models.

- **Input:** The system accepts .mp3 or .wav files via the Streamlit interface.
- **Normalization:** We use **FFmpeg** to convert all inputs to a standard format:
 - **Sample Rate:** 16,000 Hz (required by Whisper).
 - **Channels:** Mono (to reduce processing load).
 - **Bitrate:** 128 kbps.

Phase 2: Transcription (Speech-to-Text)

We utilize OpenAI's **Whisper** model (base/medium architecture) for the core transcription task.

- **Log-Mel Spectrogram:** The normalized audio is converted into a visual representation (spectrogram) of frequencies over time.
- **Encoder-Decoder Transformer:**
 - The **Encoder** analyzes the audio features.
 - The **Decoder** predicts the corresponding text tokens, handling language detection and punctuation automatically.
- **Result:** A highly accurate, time-stamped text transcript.

Phase 3: Topic Segmentation & NLP Analysis

Once we have raw text, we apply Natural Language Processing (NLP) to make it navigable.

- **TextTiling Algorithm:** We use the NLTK library to analyze "lexical cohesion." The algorithm looks for shifts in vocabulary usage (e.g., stopping the use of "weather" words and starting "sports" words) to insert boundaries where the topic changes.
- **Keyword Extraction:**

- **Stopword Removal:** Common words like "the," "is," and "and" are removed.
- **Frequency Analysis:** We calculate the most frequent meaningful nouns and verbs to represent the core themes of the podcast.

Phase 4: Visualization & Interaction

The final data is rendered into an interactive dashboard.

- **Bubble Chart:** Uses matplotlib to display keyword frequency, allowing users to spot main themes at a glance.
- **Interactive Transcript:** The segmented text is displayed in blocks, allowing users to read specific sections without scrolling through a wall of text.

3. Tech Stack

The project leverages a robust stack of open-source technologies:

- **Programming Language:** Python 3.9+
- **User Interface:** Streamlit (for the web dashboard)
- **Speech-to-Text Model:** OpenAI Whisper (Base/Small/Medium models)
- **Natural Language Processing:** NLTK (TextTiling, Stopwords, Tokenization)
- **Audio Processing:** FFmpeg, Librosa, Silero VAD.
- **Visualization:** Matplotlib, WordCloud
- **Testing Framework:** Pytest (Unit testing and Mocking)
- **Version Control:** Git & GitHub

4. Setup

Follow these steps to set up the project locally:

Prerequisites

- Python 3.8 or higher installed.
- **FFmpeg** installed and added to your system PATH.
- Git installed.

Installation

1. Clone the Repository:

```
git clone <your-github-repo-url>
```

```
cd Automated-Podcast-Transcription
```

2. Create a Virtual Environment:

Windows

```
python -m venv .venv
```

```
.venv\Scripts\activate
```

Mac/Linux

```
python3 -m venv .venv
```

```
source .venv/bin/activate
```

3. Install Dependencies:

```
pip install -r requirements.txt
```

4. Download NLTK Data: Run this python command once to download necessary NLP packages:

```
import nltk
```

```
nltk.download('punkt')
```

```
nltk.download('stopwords')
```

5. Usage

To run the application, execute the following command in your terminal:

```
streamlit run src/app.py
```

How to use the Interface:

1. Open the local URL provided (usually <http://localhost:8501>).
2. Use the sidebar to upload a podcast file (supported formats: .wav, .mp3).
3. Click the "**Process Audio**" button.
4. Wait for the pipeline to finish (Transcription -> Segmentation -> Analysis).
5. View the results:
 - **Full Transcript:** Read the complete text.
 - **Topic Segments:** See the text divided by topic changes.
 - **Keyword Analysis:** View the most discussed terms in a bubble chart.

Screenshots of UserInterface:

6. Testing

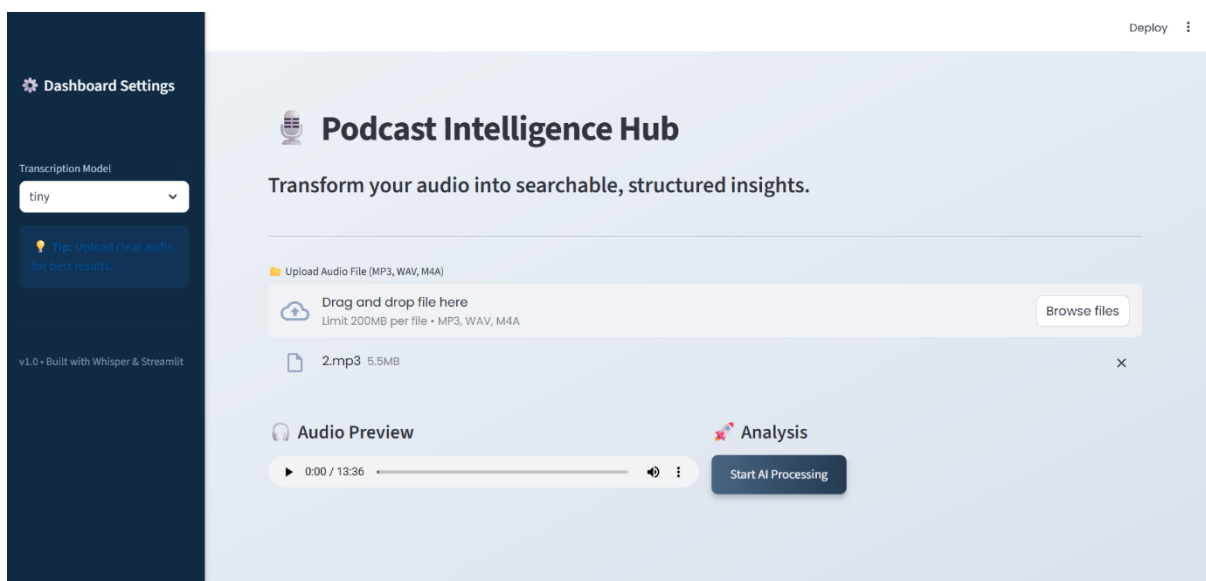
The project includes an automated testing suite to ensure reliability.

Running Tests:

```
python -m pytest
```

- **Scope:** Validates audio preprocessing logic, error handling for missing files, and the accuracy of keyword extraction algorithms.

7. User Interface



Dashboard Settings

Transcription Model

tiny

Tip: Upload clear audio for best results.

Download Full Transcript

v1.0 • Built with Whisper & Streamlit

Deploy

Keyword BubblesTopic TimelineSearch & PlayFull Transcript

Raw Text

Full Transcript

Ted Talks are recorded live at the Ted Conference and produced with WNYC New York Public Radio. This episode features MIT, Engineer, and LowTech Designer, Amy Smith. Ted Talks are made possible through the support of BMW where ideas are everything. Here's Amy Smith. Tales of invention. I'd like to tell you the tale of one of my favorite projects. I think it's one of the most exciting that I'm working on, but I think it's also the simplest. It's a project that has the potential to make a huge impact around the world. It addresses one of the biggest health issues on the planet. The number one cause of death in children under five, which is waterborne diseases, diarrhea, malnutrition. No, breathing smoke from indoor cooking fires. A cute respiratory infections caused by this. Can you believe that? I find this shocking and somewhat appalling. Can't we make cleaner burning cooking fuels? Can't we make better stoves? How is it that this can lead to over two million deaths every year? Haiti is now 98% deforested. You'll see, it seems like this all over the island. It leads to all sorts of environmental problems and problems that affect people throughout the nation. A couple years ago, there was severe flooding that led to thousands of deaths. It's directly attributable to the fact that there are no trees on the hills to stabilize the soil. So the rains come. They go down the rivers and the flooding happens. Now one of the reasons why there are so few trees is this. People need to cook and they harvest wood and they make charcoal in order to do it. It's not that people are ignorant to the environmental damage. They know perfectly well, but they have no other choice. Fossil fuels are not available. And solar energy doesn't cook the way that they like their food prepared. And so this is what they do. You'll find families like this who go out into the forest, find a tree, cut it down, and make charcoal out of it. So not surprisingly, there's a lot of effort that's been done to look at alternative cooking fuels. About four years ago, I took a team of students down to Haiti and we worked with Peace Corps volunteers there. This is one such volunteer, and this is a device that he had built in the village where he worked. And the idea was that you could take waste paper. You could compress it and make pre-cats that could be used for fuel. But this device is very slow. So our engineering students went to work on it. And with some very simple changes, they were able to triple the throughput of this device. So you could imagine they were very excited about it. And they took the Brickhead's back to MIT so that they could test them. And one of the things that they found was, they didn't burn. So it was a little discouraging to the students. And in fact, if you look closely right here, you can see it says US Peace Corps. As it turns out, there actually wasn't any waste paper in this village. And while it was a good use of government paperwork for this volunteer to bring it back with him to his village, it was 800 kilometers away. And so we thought perhaps there might be a better way to come up with an alternative cooking fuel. What we wanted to do is we wanted to make a fuel that used something that was readily available at the local level. You see there's all sorts of things well. There's small scale sugar mills. And the waste product from those after you extract the juice from

Download TXT

Dashboard Settings

Transcription Model

tiny

Tip: Upload clear audio for best results.

Download Full Transcript

v1.0 • Built with Whisper & Streamlit

Deploy

Keyword BubblesTopic TimelineSearch & PlayFull Transcript

Semantic Search

Search for a concept:
e.g. 'Future technology'

Min Sentiment Score
-1.00

Found 14 segments matching your search.

00:00 | Talks

00:57 | Make

02:10 | Could

03:06 | Sugar

04:03 | Wanted

04:43 | Cassava

05:36 | Well

06:40 | Charcoal

07:21 | Actually

08:36 | Charcoal

09:15 | Think

10:17 | Deal...

Dashboard Settings

Transcription Model

tiny

Tip: Upload clear audio for best results.

Download Full Transcript

v1.0 • Built with Whisper & Streamlit

Deploy

Audio PreviewAnalysis

0:00 / 13:36

Start AI Processing

Processing Complete!

Keyword BubblesTopic TimelineSearch & PlayFull Transcript

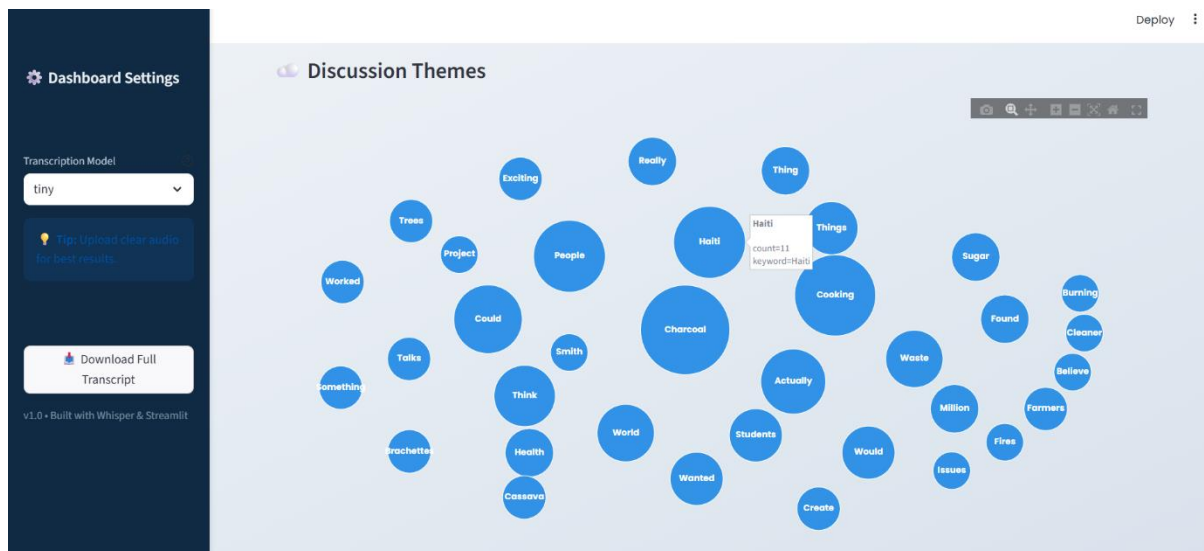
Topic Progression

Sentiment Heatmap

Topic

Timeline

Sentiment



8. Troubleshooting

- **Error: FileNotFoundError: [WinError 2] The system cannot find the file specified**
 - **Cause:** FFmpeg is not installed or it is not added to your system PATH.
 - **Fix:** Install FFmpeg, add it to your environment variables, and restart your terminal.
- **App crashes with "Out of Memory"**
 - **Cause:** The Whisper model (defaulting to "medium") is too large for your available RAM.
 - **Fix:** Open `src/transcription.py` and change the line `model_size="medium"` to `model_size="tiny"` or `model_size="base"`.
- **Error: Streamlit command not found**
 - **Cause:** The virtual environment is not currently activated.
 - **Fix:** Activate your environment before running the command:
 - **Windows:** `.venv\Scripts\activate`
 - **Mac/Linux:** `source .venv/bin/activate`

Changes made:

1. **Main Bullets:** Used for the Error messages so they stand out.
2. **Sub-bullets:** Indented "Cause" and "Fix" to keep them organized under the specific error.
3. **Code formatting:** Added backticks (`````) around commands and file paths (like `src/transcription.py`) to make them look like code.

9. Limitations and Challenges

- **Processing Speed:** Transcribing long audio files (1 hour+) can take significant time on a CPU. GPU acceleration is recommended for faster performance.
- **Overlapping Speech:** The current model struggles to distinguish individual speakers when multiple people talk at once (Speaker Diarization is not yet implemented).
- **Audio Quality:** Heavy background noise or low-quality recording equipment significantly impacts transcription accuracy.

10. Future Scope

- **Speaker Diarization:** Implement pyannote.audio to identify *who* is speaking (e.g., "Speaker A", "Speaker B").
- **Real-Time Transcription:** Adapt the pipeline to transcribe live audio streams rather than just uploaded files.
- **Summarization:** Integrate a Large Language Model (like GPT or BART) to generate a concise 3-paragraph summary of the entire episode.
- **Cloud Deployment:** Deploy the application to AWS or Streamlit Cloud for public access.

11. References

1. **OpenAI Whisper:** <https://github.com/openai/whisper>
2. **Streamlit Documentation:** <https://docs.streamlit.io/>
3. **NLTK TextTiling:** <https://www.nltk.org/api/nltk.tokenize.html>
4. **FFmpeg:** <https://ffmpeg.org/>